

ASIM ASHFAQ

Co-Founder & CTO | Full Stack | Distributed Systems & Cryptography

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PROFILE

Software engineer with 12+ years across full stack product work, distributed systems and applied cryptography. Currently Co-Founder & CTO at [OxEquity](#), where I built and run the platform end to end: a NestJS backend of some fifty domain modules, a React 19 dApp, a Go agentic execution stack, and the zero-trust infrastructure all three sit on.

Recent work has been in agent tooling and zero-knowledge proving: an MCP server that lets a language model author and run multi-step DeFi strategies under keys that cannot move funds, and a Groth16 proving stack in Rust.

EXPERIENCE

Founder

DeFiLoops | Remote | 2026 – Present

A multi-step DeFi agentic workflow platform: you describe a strategy in one sentence, it becomes a numbered plan you approve, and it then runs unattended across several networks, on an account only you can empty.

- Built the Go execution stack: a six-stage compiler from conversation to signed transaction, where each stage narrows what is possible and any stage can refuse.
- Designed the key model so it holds on-chain rather than by policy: the agent signs typed adapter parameters with no field for a destination or raw calldata, is capped by a spend limit it cannot raise, and is revocable across a whole chain in one transaction.
- Wrote the MCP server that lets a language model author, validate and run these plans, with the money-spending tools withheld at the server rather than by convention.
- Shipped the adapter set behind it (swaps, lending, liquidity, leverage, bridging and property) with a router that answers what actually fills at a named limit price.
- Durable runs on Temporal, plus schedules, price triggers and plan chaining, so a strategy left on for months is rebuilt and re-priced each time it fires.
- Wrote the public documentation and the 21 agent skill guides the model reads before it answers.

[Go](#) · [MCP](#) · [Temporal](#) · [Solidity](#) · [Foundry](#) · [ERC-4337](#) · [PostgreSQL](#)

Co-Founder & CTO

OxEquity US, Inc. | Remote | Nov 2022 – Present

- Founded and built the engineering side of a real-world-asset tokenisation platform: tokenised real estate and equities, traded and used as collateral across Base, Arbitrum, Ethereum and Unit0.
- Wrote and maintain the NestJS/TypeScript backend: roughly fifty domain modules covering KYC/AML, orders, trades, a match-settlement engine, holdings, rewards, fiat on/off-ramp, campaigns and relayed on-chain transfers.
- Designed the platform's zero-trust infrastructure: SPIFFE/SPIRE workload identity issuing mTLS to every service, HashiCorp Vault for secrets and signing keys, Falco for runtime detection, and immudb as a tamper-evident audit ledger.
- Moved long-running, externally-settled operations onto Temporal, so cross-chain bridge steps, unfilled limit orders and transactions needing fee bumps survive restarts instead of being lost mid-flight.
- Built authentication around passkeys and OPAQUE rather than transmitted passwords, with Casbin policy for authorisation and TOTP as a second factor.
- Built the React 19 dApp on wagmi/viem (trading, portfolio, lending and a pro order ticket) with a swap router that reports real price impact rather than an unmeasured zero.
- Run the whole delivery loop solo: reproducible Nix dev shells, a Dockerised local stack, CI, and Prometheus/Grafana/Metabase for what happens after deploy.

[NestJS](#) · [TypeScript](#) · [Go](#) · [React 19](#) · [Temporal](#) · [PostgreSQL](#) · [SPIFFE/SPIRE](#) · [Vault](#) · [Solidity](#)

Full Stack Developer

Jarvis OOD | Remote | Mar 2020 – Present

- Front-end work on Jarvis Network, a set of protocols connecting traditional finance to DeFi. Synthetium issues synthetic fiat assets (jEUR, jCHF and other jFIATs) priced by Chainlink oracles, governed by the Jarvis DAO and its JRT token.
- Built the platform's interfaces in React and Next.js.

[React](#) · [Next.js](#)

DevOps / Full Stack Developer

EverReal GmbH | Remote | Oct 2020 – Aug 2021

- Worked on EverReal, a Munich proptech running a cloud-based white-label platform that digitises the whole rental and sales transaction: listing and marketing, applicant management, viewing appointments, electronic contracts and handover.

- Split between application development and the infrastructure the B2B SaaS platform ran on.

[SaaS](#) · [DevOps](#)

DevOps / Full Stack Developer

WeFresh, Ascend Commerce | Remote | Dec 2019 – Aug 2020

- Worked on WeFresh, Ascend Commerce's Bangkok grocery and convenience delivery app, which fulfils orders through 7-Eleven, CP Freshmart and Makro wholesale.
- Split between backend service work and the delivery and deployment infrastructure behind it.

[Node.js](#) · [DevOps](#)

Team Lead

Expanse.tech | Islamabad | Aug 2016 – 2019

- Led the team delivering Gander, Tokenlab and Luna Wallet across the Expanse blockchain ecosystem.

[Vue](#) · [React](#) · [Golang](#) · [Electron](#) · [Solidity](#)

Senior .NET Developer / Project Lead

Zigron (Pvt) Ltd. | Islamabad | Mar 2016 – May 2016

[.NET](#) · [C#](#) · [SQL Server](#)

.NET Developer

Defence Housing Authority | Islamabad | Dec 2015 – Mar 2016

[.NET](#) · [C#](#)

Freelance Developer

Freelancer.com | Islamabad | 2012 – Mar 2017

- Web and software development for clients on Freelancer.com, working there since 2012: WordPress and PHP/MySQL builds, e-commerce and payment gateway integration, and third-party API work across PayPal, Facebook and Google.
- Rated 4.7 out of 5 across 72 reviews, with 88% on-time delivery, 94% on-budget completion, and 39% of clients hiring again.
- Larger engagements included a SaaS clinic management system (Angular 2, MEAN) and a realtime chat/video application (Node.js, MQTT, STOMP, RabbitMQ).

[Angular 2](#) · [MEAN](#) · [Node.js](#) · [MQTT](#) · [RabbitMQ](#)

PROJECTS

OxEquity (Oxequity.com)

2022 – present

A platform for investing in tokenised real-world assets (property and equities) and putting them to work on-chain. Fractional positions from small tickets, rental yield paid on-chain, and holdings usable as collateral while they keep earning.

- NestJS backend of ~50 domain modules, from KYC and document signing through orders, settlement, holdings and rewards.
- Multi-chain across Base, Arbitrum, Ethereum and Unit0, with an RPC proxy that spends a read's retries on different providers rather than failing on one.
- Zero-trust service mesh: SPIRE-issued SPIFFE identities, mTLS between every service, Vault-held signing keys, immudb audit ledger, Falco runtime detection.
- Durable Temporal workflows for anything that settles outside a request: bridge legs, resting orders, transaction bumps.
- React 19 dApp on wagmi/viem with RainbowKit, TanStack Query, Zustand and i18n.

[NestJS](#) · [TypeScript](#) · [React 19](#) · [wagmi/viem](#) · [Temporal](#) · [PostgreSQL](#) · [TypeORM](#) · [Redis](#) · [RabbitMQ](#) · [MQTT](#) · [gRPC](#) · [SPIFFE/SPIRE](#) · [Vault](#) · [Falco](#) · [immudb](#) · [Casbin](#) · [WebAuthn](#) · [Docker](#) · [Nix](#)

DeFiLoops (defiloops.com)

Go | agentic execution

The execution layer for multi-step DeFi. You describe a strategy in a sentence; it becomes a numbered plan you approve, and then runs for hours or days across networks, on an account only you can empty.

- Six-stage compiler from conversation to signed transaction, where each stage narrows what is possible and any stage can refuse.
- Steps that belong together are merged into one transaction, so an approval cannot outlive the step it was granted for; a step can spend what the previous step produced.
- Atomic loan migration via flash loan: a deadlock by hand, one instruction here, with the borrowed funds never touching the user's account.
- Key model enforced on-chain rather than by policy: the agent key signs typed adapter parameters with no field for a destination or raw calldata, is capped by a spend limit it cannot raise, and is revocable for every account on a chain in one transaction.
- Own swap router answering questions an aggregator cannot: what fills at a named limit price, and how much a resting order takes right now.
- Durable runs on Temporal, plus schedules, price triggers and plan chaining that rebuild and re-price the plan each time they fire.
- Foundry invariant tests for the adversarial cases: agent blast radius, guard drain, proxy squatting, address parity.

[Go](#) · [Temporal](#) · [Solidity](#) · [Foundry](#) · [ERC-4337](#) · [PostgreSQL](#) · [go-ethereum](#) · [Uniswap v3](#) · [Aave](#) · [CCTP](#)

MCP server for agent-driven DeFi

part of DeFiLoops

A Model Context Protocol server that lets a language model author, validate and run multi-step on-chain strategies, built to find out whether a small open model could write a plan the system would accept, and to make the answer yes.

- 22 tools in five groups (catalogue, chain reads, plan authoring, chains and schedules, and run watching) over ~33k lines of Go with 84 test files.
- Two transports with different guarantees: stdio, where a pipe's single reader makes unexposed tools structurally unreachable; and a loopback socket requiring a bearer token, refusing to bind anything but localhost.
- Tools that spend money are withheld at the server unless explicitly enabled. On a pipe the client's request is a guarantee; on a socket it is only a preference.
- Validation goes over the wire to the running API rather than importing the compiler, so the model validates against the build that will actually execute it.
- Per-request credential binding on a value receiver, so two concurrent conversations cannot swap identities under load.
- 21 written skill guides the agent reads before answering, held to one rule: a guide describes what the code does today, or it starts lying.

MCP · Go · JSON-RPC 2.0 · LLM tool use · OpenRouter

rapidzkey (github.com/asimashfaq/rapidzkey)

Open source | MIT

Fast Groth16 trusted setup for Circom circuits, a drop-in replacement for `snarkjs groth16 setup` producing byte-identical output.

- 20–60× faster than snarkjs: a ~2M-constraint circuit drops from over an hour to about three minutes on eight cores.
- Accumulates weighted curve points by scatter-add across Rayon threads, rather than running a separate MSM per signal.
- Computes missing Lagrange-basis points by parallel IFFT, so unprepared ptau files work without a separate preparation pass.

Rust · Rayon · arkworks · Groth16 · BN254 · Circom

zk-realestate

Rust | zero-knowledge

Rust · arkworks · Circom · WASM · napi-rs · axum · BLS · WebAuthn

Content Engine

Python | Dagster | 0xEquity

Python · Dagster · Claude API · MCP · FastAPI · PostgreSQL · Whisper · Vault · Nix

indexer-core

TypeScript / Bun | Bazel

TypeScript · Bun · Bazel · Nix · Dagster · PostgreSQL · Foundry

Jarvis Network (jarvis.money)

2020 – present

React · Next.js · Solidity · Chainlink · Ethereum · Polygon · Synthetic assets · AMM routing

Full Stack Starter (github.com/asimashfaq/fullstack-starter)

Open source

NestJS · Next.js · gRPC · GraphQL · nx · TypeScript · RavenDB · Storybook · Envoy · Docker

Gander (gander.tech)

Expanse.tech | 2016 – 2019

Golang · React · Redux · Elasticsearch · Couchbase · MongoDB · Redis · WebSockets · JSON-RPC · AWS

Tokenlab (tokenlab.io)

Expanse.tech | 2016 – 2019

Vue · Golang · Node.js · Solidity · Truffle · Ganache · Mocha · web3.js · HD Wallet · Meta transactions

Luna Wallet (github.com/expanse-org/luna-wallet)

Expanse.tech | Open source

Vue · Electron · Blockchain node · ABI · REST API

Other earlier work

Clinic Management System: SaaS on Angular 2 and the MEAN stack. · **Random Chat / Video App:** Node.js, MQTT, STOMP, RabbitMQ. · **Places App** and **TicTac App:** React, on GitHub.

SKILLS

AI tooling

Model Context Protocol · LLM tool-use design · Agent skill authoring · Claude API

Languages

TypeScript / JavaScript · Go · Rust · Python · Solidity · SQL

Backend

NestJS / Node.js · Temporal (durable workflows) · Dagster (data orchestration) · REST / GraphQL / gRPC · RabbitMQ / Kafka / MQTT · WebSocket / realtime PubSub · Microservices

Frontend	React / Next.js · wagmi / viem / RainbowKit · TanStack Query / Zustand · Tailwind CSS · Vue / Angular / Electron
Data	PostgreSQL / TypeORM · Redis · NoSQL / Elasticsearch · immudb / Hasura
Blockchain & Web3	ERC-4337 account abstraction · Foundry / Hardhat · EIP-2535 Diamonds · DeFi: AMMs, lending, flash loans · Cross-chain bridging (CCTP) · RWA tokenisation
Cryptography	Groth16 zkSNARKs / arkworks · Circom circuits · WebAuthn / passkeys · OPAQUE / BLS signatures
Infrastructure	Kubernetes / k3s / Rancher · Docker / Nix / Helm / Istio · Bazel / Bun · Terraform / GitOps · SPIFFE / SPIRE / Vault · Falco / Open Policy Agent · Prometheus / Grafana · GitLab CI / CircleCI

EDUCATION

BS Software Engineering

Riphah International University, Islamabad
2010 – 2014

I.Com

Punjab College of Commerce, Rawalpindi
2005 – 2007

HSSC

Fazaia International School, Rawalpindi
2003 – 2004

VOLUNTEERING

Founder, “Pukaray Pakistan”

Social organisation, Islamabad
2009 – 2015

Volunteer, IEEE Pakistan Mission to Progress

IEEE Pakistan Women in Engineering Forum,
Lahore
2012 – 2013

Volunteer, Aghosh Foundation

Attock
2009 – 2010

LANGUAGES

English (Professional) · Urdu (Native)

QUOTE

“Shoot for the moon. Even if you fail, you will fly among the stars.”



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